

HINAN BILAL

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PROFESSIONAL SUMMARY

Senior Java Full Stack Engineer with 5+ years of experience building scalable, high-performance enterprise systems using Java, Spring Boot, and Angular. Expertise in microservices architecture, event-driven systems with Kafka, and cloud-native solutions on AWS. Proven track record of designing systems handling large-scale data (1TB+ logs) and high user traffic across banking, telecom, and government domains. Strong experience in distributed systems, system design, and leading engineering teams. Additional expertise integrating machine learning models into production Java systems.

SKILLS

Languages	Java, Python, SQL
Backend	Spring Boot, Microservices, REST APIs, Hibernate, JPA, Maven/Gradle
Security & Testing	Spring Security, JWT/OAuth2, Swagger/OpenAPI, JUnit/Mockito
Frontend	Angular, TypeScript
Messaging & Streaming	Kafka
Databases	PostgreSQL, Oracle, Redis
Cloud & DevOps	AWS (Lambda, S3, Step Functions, CloudFormation), Docker, Jenkins, Git/GitHub
Monitoring & Logging	ELK Stack, SonarQube
AI/ML Integration	Gen AI, Scikit-learn, ML model integration with Java services, Python
Architecture	Distributed Systems, Event-Driven Architecture, Scalable Systems Design, Agile/Scrum

EXPERIENCE

Senior Software Engineer Confiz Ltd.	Jun 2025 - Present Islamabad, Pakistan
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- Served as a **Line Manager and Senior Engineer**, managing and mentoring engineers, defining KPIs, conducting performance evaluations, and ensuring high-quality and timely delivery across multiple teams.
- Contributed to development of enterprise-grade government systems, participating in key architectural decisions and ensuring scalable, high-quality implementations.
- Designed and deployed a centralized **ELK stack processing 1TB+ logs**, integrated with IRIS platform, Provincial Revenue Authority systems, and Digital Invoicing System, significantly improving observability and log analytics across distributed services.
- Contributed to backend development and architectural design of **Provincial Revenue Authority systems**, supporting large-scale digital transformation using Java and Spring Boot-based microservices.
- Led design and implementation of a **Fleet Management System**, handling vehicle lifecycle, assignments, transfers, and multi-level approval workflows across large organizations.
- Led development of a **Third-Party Auditor Management System**, enabling task assignment, workflow automation, performance evaluation, and incentive processing for external auditors and contractors.
- Mentored engineers through code reviews, technical guidance, and best practices, improving team productivity and system maintainability. Conducted 50+ technical interviews.
- Tech Stack:** Java 21, Spring Boot, Microservices, Kafka, ELK Stack, Redis, WSO2, SonarQube

Senior Software Engineer Systems Ltd.	Sep 2021 - Jun 2025 Islamabad, Pakistan
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- Developed microservices-based **Etisalat B2B platform**, handling organization document workflows, expiry tracking, and customer portfolio aggregation; implemented OCR-based validation and integrated TIBCO EDMS for secure document storage.

- Designed and built scalable backend services using **Java, Spring Boot, and REST APIs**, enabling seamless inter-service communication via Feign clients and ensuring reliable transaction logging.
- Contributed to development of **retail banking platforms** serving **375+ branches**, building high-availability backend services using **Java, Spring Boot, and Temenos**.
 - Delivered core financial modules including **RAAST P2P/P2M, IBFT, FT, utility bill payments, telecom services, wallet top-up, card management, and billing favourites**.
 - Integrated **interbank payment networks and external services** for real-time financial transactions.
 - Built notification systems supporting **email, SMS, and push notifications** for transaction alerts.
- Built backend components for a high-traffic **US-based media platform (150M+ monthly users)**, optimizing content delivery via CDN, implementing Firebase push notifications, and improving content tagging workflows.
- Contributed to **digital transformation of a Dubai government department**, developing microservices using Java 17 and Spring Boot.
- Developed **telecom reload system PoC** using microservices architecture and AWS, implementing authentication, balance validation, and transaction workflows.
- Supported cloud infrastructure for a **Fortune 500 biotech client**, utilizing AWS services including Lambda, Step Functions, S3, and CloudFormation.
- Integrated an **ML clustering model into a Spring Boot service** via MQTT for real-time driver fatigue detection, triggering Firebase push notifications to proactively prevent traffic incidents.
- **Tech Stack:** Java 17, Spring Boot, Microservices, AWS, Kafka, Firebase, SQL

Full Stack & ML Engineer
Freelance

Jan 2020 - Sep 2021
Islamabad, Pakistan

- Delivered 50+ backend and data-driven projects using **Java, Spring Boot, and AWS**, building scalable REST APIs and microservices-based systems.
- Migrated monolithic Spring Boot applications to **microservices architecture**, implementing API Gateway, service discovery, centralized configuration, and inter-service communication using Feign clients and SQS.
- Designed backend systems with **PostgreSQL and Redis**, improving performance through caching strategies and optimized database queries.
- Led a team of 3 developers, managing task distribution, code reviews, and ensuring timely and high-quality project delivery.
- Applied machine learning techniques (regression, clustering) in selected projects for data analysis and predictive insights.

PROJECTS

PakistanTravelAdvisor.com

Python, BeautifulSoup, ML

- Built a full-stack travel platform with a search engine, hotel aggregator, and ML-based itinerary planner.
- Automated bi-daily data ingestion via BeautifulSoup cron jobs across multiple travel providers.
- Grew to **10,000+ site visits** and **7,000+ community members**; acquired by a Swedish entrepreneur in 2022. ([Watch demo](#))

Employee Service Portal

Vue.js, AWS Amplify, CloudFormation

- Developed a corporate portal supporting authentication, working hours tracking, and leave management workflows.
- Deployed end-to-end on AWS using CloudFormation and Amplify for scalable, serverless hosting.

Predictcoronavirus.tech

Python, Machine Learning

- Deployed an ML-based web app predicting Pakistan's COVID-19 trajectory 30 days ahead.
- Successfully forecast the curve flattening significantly ahead of official projections. ([Visit archive](#))

twitterEye — Python Package

Python, NLP, PyPI

- Published an open-source PyPI package for analyzing and visualizing public sentiment on Twitter topics.
- Enabled keyword-level reaction tracking and trend visualization for researchers and developers. ([Visit PyPI](#))

EDUCATION

Bachelor of Computer Science, NAMAL University

Relevant Coursework: Data Structures, Databases, Computer Networks, and Machine Learning.

CERTIFICATIONS

Amazon Web Services Solutions Architect - Associate

Issued: Dec 2025

Amazon Web Services Cloud Practitioner

Issued: Nov 2022